LAB ASSIGNMENT-6.5

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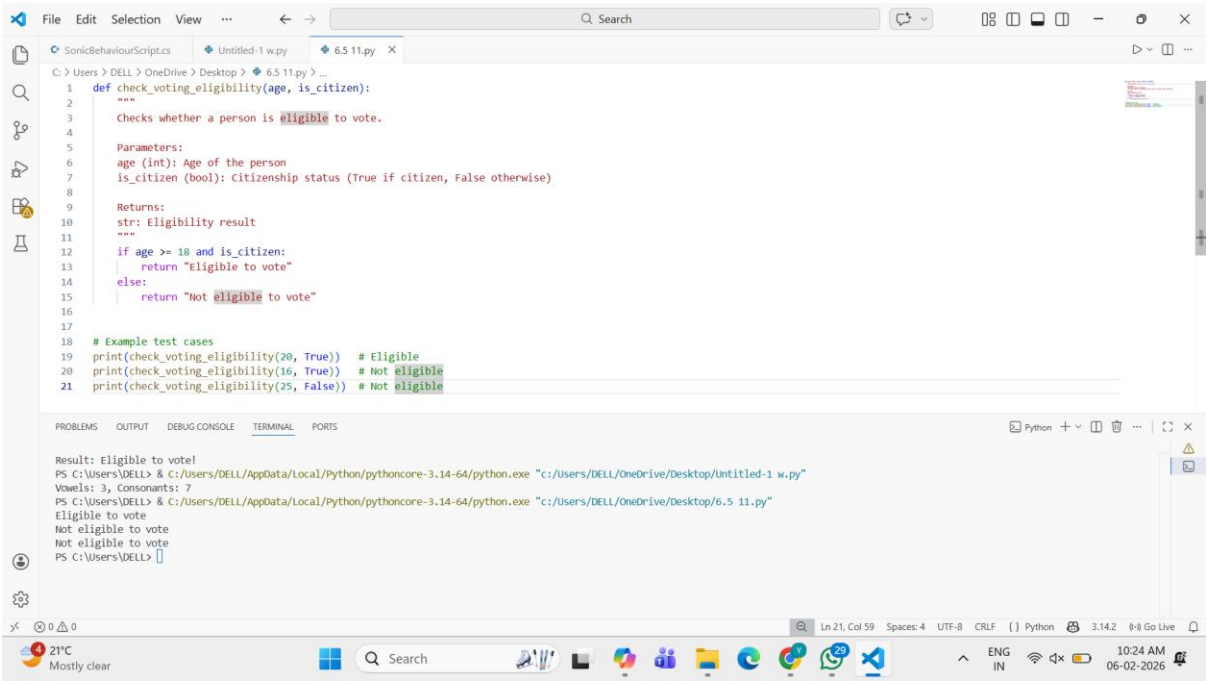
TASK-01: (AI-Based Code Completion for Conditional Eligibility Check)

PROMPT: "Generate Python code to check voting eligibility based on age and citizenship."

CODE AND OUTPUT :



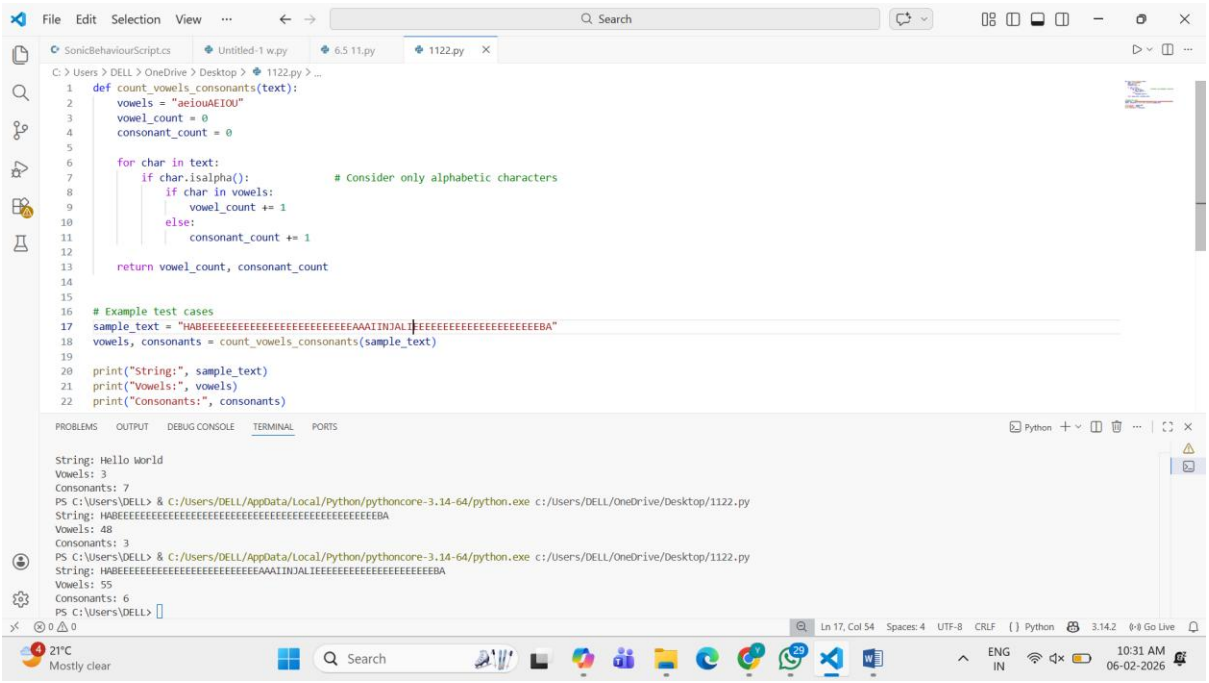
OBSERVATION: The AI-generated code effectively uses conditional statements to check voting eligibility by verifying both age and citizenship. By applying the logical AND operator, the program ensures that a person is eligible

to vote only when all required conditions are satisfied. The logic is clear, accurate, and easy to understand, leading to correct eligibility decisions for different input cases.

TASK-02: (AI-Based Code Completion for Loop-Based String Processing)

PROMPT: "Generate Python code to count vowels and consonants in a string using a loop."

CODE AND OUTPUT:



OBSERVATION:

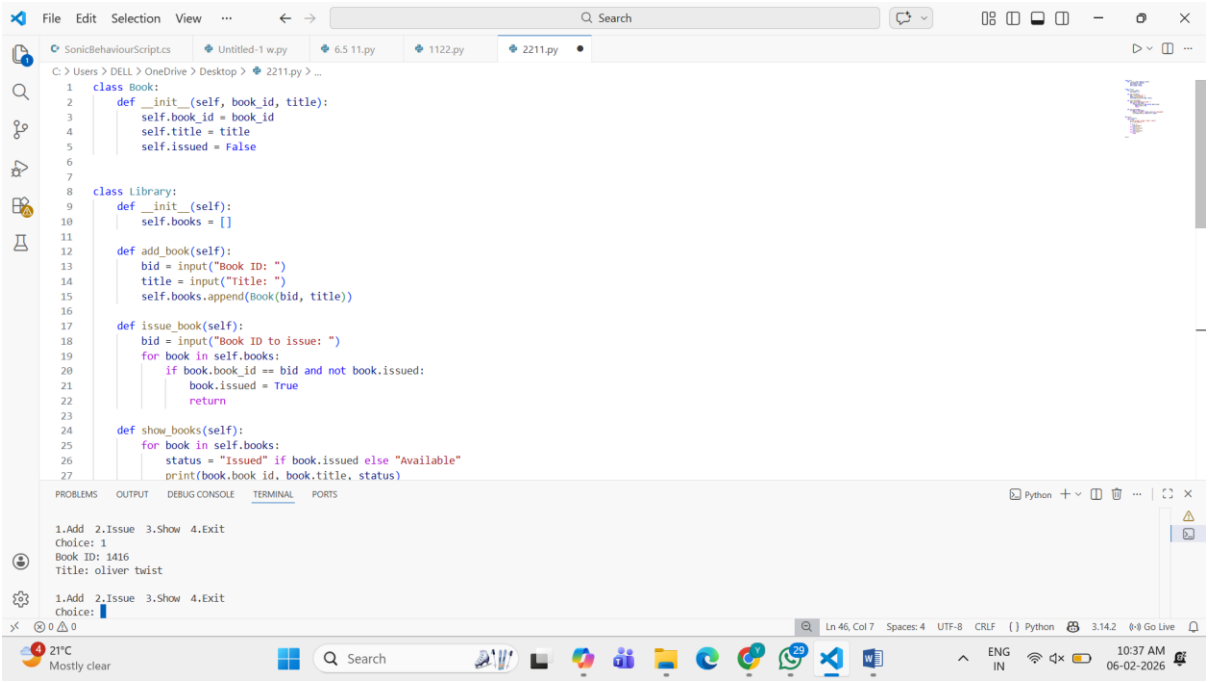
The AI-generated code efficiently processes a string using a loop to count vowels and consonants. It correctly iterates through each character, checks whether it is an alphabet, and classifies it as a vowel or consonant using conditional logic. The approach produces accurate counts while ignoring non-alphabetic characters, ensuring reliable output. Overall, the solution is clear,

logically structured, and demonstrates effective use of loops and conditions for string processing.

TASK-03: (AI-Assisted Code Completion Reflection Task)

PROMPT: "Generate a Python program for a library management system using classes, loops, and conditional statements."

CODE AND OUTPUT:



OBSERVATION:

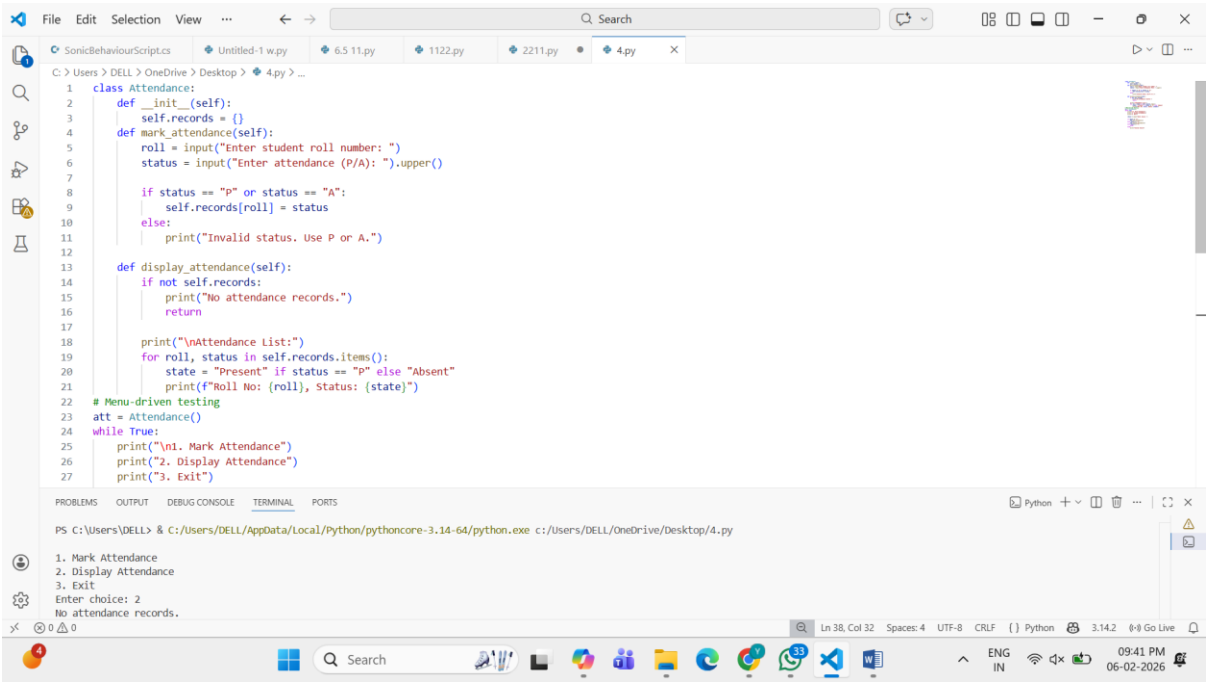
The AI-generated library management program effectively integrates classes, loops, and conditional statements to create a functional, menu-driven system. The class structure is clear and well-organized, making the code easy to understand and maintain. The use of loops enables continuous user interaction, while conditionals correctly handle different operations such as adding, issuing,

and displaying books. Overall, the AI suggestions are logically sound, practical, and provide a strong foundation that can be easily extended or improved with additional features.

TASK-04: (AI-Assisted Code Completion for Class-Based Attendance System)

PROMPT: "Generate a Python class to mark and display student attendance using loops."

CODE AND OUTPUT:



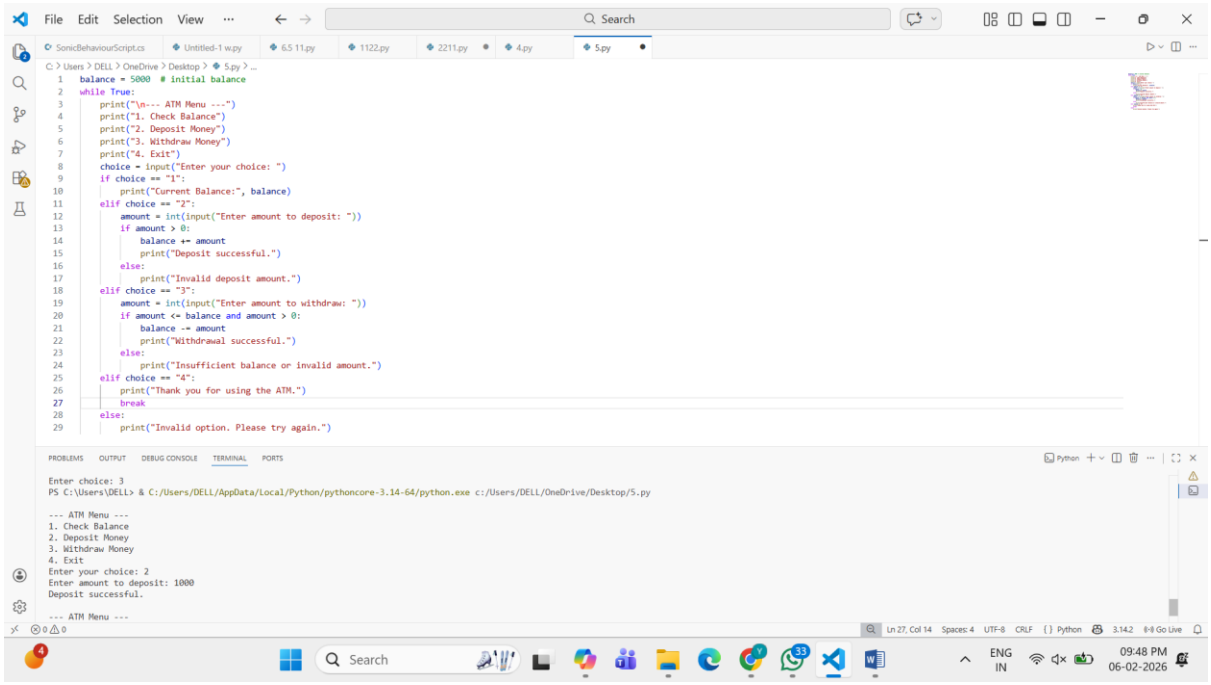
OBSERVATION:

The AI-generated attendance system correctly uses a class to store and manage student attendance records. Loops are effectively applied to iterate through records for displaying attendance, while conditional statements ensure valid marking of present or absent status. The logic is simple, accurate, and produces correct attendance displays for different test cases. Overall, the AI provides a clear and reliable solution that demonstrates proper use of object-oriented concepts along with loops and conditionals.

TASK-05: (AI-Based Code Completion for Conditional Menu Navigation)

PROMPT: "Generate a Python program using loops and conditionals to simulate an ATM menu."

CODE AND OUTPUT:



OBSERVATION:

The AI-generated ATM program effectively uses loops and conditional statements to simulate menu-based navigation. The loop ensures continuous user interaction, while conditional logic correctly handles options such as checking balance, depositing money, withdrawing money, and exiting the system. Input validation helps prevent invalid transactions, leading to accurate and reliable outputs. Overall, the AI solution is clear, well-structured, and successfully demonstrates conditional menu navigation.